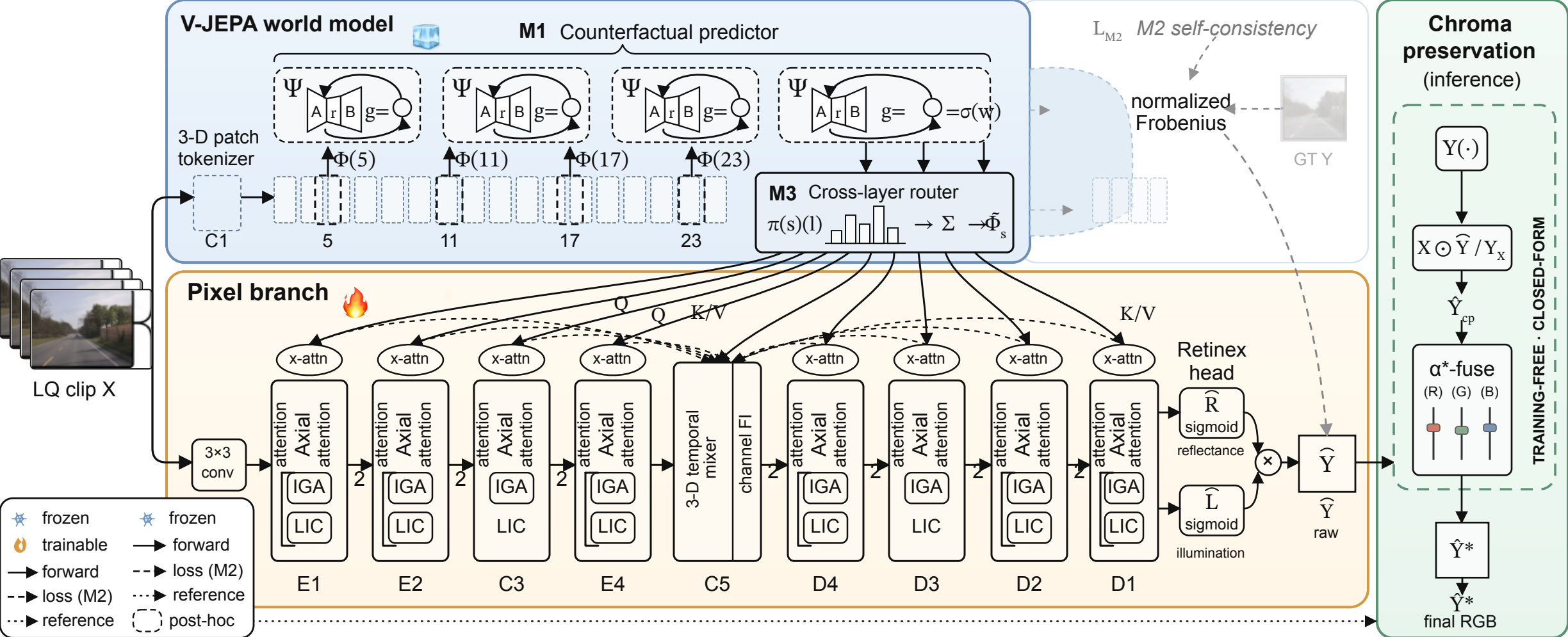

V-JEPA world model
M1 Counterfactual predictor
3-D patch tokenizer
C1
Φ(5)
Φ(11)
Φ(17)
Φ(23)
5
11
17
23
M3 Cross-layer router
π(s)(l)
→ Σ →Φ̃s
L_M2 M2 self-consistency
normalized Frobenius
GT Y
Chroma preservation
(inference)
Y(·)
X ⊙ Ŷ / Y_X
Ŷ_cp
α*-fuse
(R) (G) (B)
Ŷ*
final RGB
TRAINING-FREE · CLOSED-FORM
Pixel branch
LQ clip X
3×3 conv
x-attn
Q
K/V
attention
Axial attention
IGA
LIC
2
3-D temporal mixer
channel FI
Retinex head
R̂ sigmoid
reflectance
L̂ sigmoid
illumination
Ŷ raw
E1
E2
C3
E4
C5
D4
D3
D2
D1
frozen
trainable
forward
loss (M2)
reference
frozen
forward
loss (M2)
reference
post-hoc